

core/utils.py::build_model_registry() scans nine domain apps; every generic endpoint is driven by the registries

Django Vector Raster WMS

DOMAIN APPS (ALLOWED_APPS) — MODELS · CALCULATIONS.PY · VIEWS

administrative 5
Province > City >
District > Neighborhood
PopulationProjection

physicalEnv 12
LandCover vec + raster
DEM · DSM · imagery
materials, env. costs

watersupply 17
extraction · treatment
pipes · NRW · OPEX
SupplySecurity

housing 8
supply/demand · HPI
mortgage · rentals
affordability

urban_heat 11
UTCI · PET · Tmrt
LST · SVF · SUHII
NBS · WBGT

builtup 6
ZoningArea · Street
Park · Facility
Building · Property

nature 6
green / blue
layers

weather 4
WMS layers
time steps

Energy 2
energy labels

Number tags = model classes per app. Indicator dashboards (watersupply, housing, urban_heat) follow calculations.py → views._build_indicators() → HTMX partial; MOCK_DATA fills in when a province is missing.

DAG edges documented in calculations.py docstrings → core/DAG.dot (DPSIR)

scan models

build_model_registry()
core/utils.py — scans allowed_apps at startup

has GeometryField, no RasterField → VECTOR
has RasterField → RASTER · 'WMS' in key → WMS

MODEL_REGISTRY

all models

VECTOR_REGISTRY

GeoJSON, bounds, editing

RASTER_REGISTRY

COG export, tile URLs

WMS_REGISTRY

WMS layer catalog

importer

field mapping, upsert keys

mainMap

/api/layers/, GeoJSON, editing

core.signals

post_save → COG, cache bump

weather

WMS catalog, tile proxy